

Attention-Based BiLSTM for Multi-Target Renewable Energy and Load Forecasting: A Fair Multi-Model, Multi-Dataset Study

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Abstract: Accurate short-term forecasting of renewable generation and electrical load is essential for the reliable operation of modern power systems with high penetration of variable resources. This paper presents a unified, multi-model and multi-dataset benchmarking study of three recurrent architectures—SimpleLSTM, bidirectional LSTM (BiLSTM), and an AttentionBiLSTM with Bahdanau-style additive attention—for joint forecasting of solar photovoltaic output, wind power, and grid load demand. All models are tested with the same experimental setup: consistent preprocessing, 24-hour input window, 12-hour forecast horizon, weighted MSE–MAE loss, and 5-fold rolling-origin cross-validation. The experiments use two real datasets: a local microgrid dataset and the German Open Power System Data (OPSD) dataset. performance evaluated with RMSE, nRMSE, MAPE, sMAPE, and check statistical significance using paired t-tests on the fold-wise RMSE values. The results show that AttentionBiLSTM gives better stability on some targets, especially local load forecasting, while the simpler models perform just as well in other cases. In particular, SimpleLSTM obtains a significantly lower RMSE for OPSD wind power ($p = 0.028$) and very good accuracy for national-scale load forecasting, with sMAPE reaching 2.05%. In summary, this work offers a fair and repeatable comparison that shows the real strengths and limits of attention mechanisms for multi-target energy forecasting.

Keywords: Renewable Energy Forecasting · Deep Learning · Attention Mechanism · Multi-target Forecasting

1 Introduction

As more solar and wind energy gets integrated into power grids, these systems have changed a lot, making it really important to have good predictions for what’s coming in the near future to keep everything stable and running smoothly [1,2]. Machine learning plays a big role here because it can pick up on trends from past records and help operators make smarter choices based on solid info. Older machine learning methods like ARIMA, smoothing techniques, support vector machines, random forests, and boosting algorithms have been used a bunch for predicting time-based data, but they often fall short when dealing with the wild ups and downs in renewable energy patterns that aren’t steady or linear [9]. Solar panels and wind turbines rely heavily on the weather, which means their

output can swing a lot in short periods [3, 4]. This adds a ton of unpredictability for those running the grid, messing with things like planning which units to use, how to distribute power economically, and keeping balance in real time, especially as more renewables come online [5, 6]. That’s why having spot-on short-term estimates for solar production, wind generation, and overall electricity demand is key to staying reliable and cutting down on costs [7, 8]. Lately, deep learning approaches, especially those using recurrent networks and long short-term memory setups, have shown they’re great at handling tricky time patterns [10, 11]. Bidirectional versions of these LSTMs boost how well they represent data by looking at sequences from both ends, and adding attention helps zero in on the most important moments in those uneven renewable flows [10, 12].

Despite these advances, several key limitations persist in the literature. Many studies address only a single forecasting task and rely on a single dataset, limiting generalisation across spatial scales and operational conditions [13, 14]. Furthermore, inconsistencies in preprocessing pipelines, feature engineering approaches, loss functions, and evaluation metrics make it difficult to draw fair conclusions about architectural improvements [15, 16].

To address these limitations, this paper presents a unified multi-output forecasting framework based on an attention-enhanced bidirectional LSTM (AttentionBiLSTM). The model jointly predicts solar PV power, wind power, and electrical load over a 12-hour horizon using the preceding 24 hours of data. The proposed approach is systematically compared with two strong baselines—a stacked BiLSTM and a standard LSTM—under identical preprocessing, feature sets, training configurations, and evaluation procedures [17]. Experiments are conducted on two real-world datasets of different scales: an in-house microgrid dataset and the Open Power System Data (OPSD) Germany dataset [18]. Performance is evaluated using RMSE, normalized RMSE, MAPE, sMAPE, and RMSE standard deviation, with statistical significance assessed via paired t -tests [19].

2 Related Work

Early studies on short-term energy forecasting relied primarily on statistical and conventional machine learning approaches such as ARIMA, exponential smoothing, support vector regression, and random forests [9, 20]. While these older methods can do okay when the data is pretty steady and close to linear, they usually have trouble picking up on the big nonlinear patterns, quick jumps in values, and different time scales that show up in renewable power output and electricity demand [5]. To handle the changing nature of the data (non-stationarity), people have tried mixing in decomposition tricks like EEMD or CEEMDAN with learning models. That helps smooth things out, but it adds extra steps in getting the data ready and makes the whole setup more sensitive to small changes [21, 22]. As deep learning has gotten better, recurrent neural networks—especially long short-term memory (LSTM) models—have become really popular for energy predictions. They’re good at grabbing long-term patterns in time data without needing someone to manually pick lag times [10, 11]. Bidirectional LSTMs take it

further by looking at both past and future context in the same layer, which gives better results for forecasting load, solar, and wind [10,21,23]. Lately, attention mechanisms have been added to these recurrent setups to focus more on the time steps or features that matter most at any moment [12]. Hybrid models with attention—sometimes mixed with convolutions, Transformers, or even transfer learning—have shown solid performance in short-term load forecasting and handling multiple energy scenarios [24,25]. That said, these more advanced models usually bring higher complexity, need a lot of fine-tuning of hyperparameters, and don’t always work equally well across different datasets or locations [14].

2.1 Identified Research Gap

Despite the encouraging progress in the field, quite a few gaps still exist in what’s been published so far. Most papers stick to predicting just one thing at a time—like only solar output, or only wind, or only load—instead of looking at solar, wind, and load together in the same model [13]. Also, very few studies test their models across completely different datasets, which makes it hard to really know how well the approach holds up in new places or under varied conditions [14]. Furthermore, reported architectural improvements are often confounded by differences in preprocessing pipelines, feature engineering strategies, or loss functions, making fair comparisons difficult [15]. Statistical significance testing of performance differences is also seldom conducted [19]. This study addresses these gaps by presenting a unified, multi-target, multi-dataset comparison of SimpleLSTM, BiLSTM, and AttentionBiLSTM models under a consistent experimental protocol.

3 Methodology

This section describes the complete experimental pipeline and the design choices adopted to ensure a fair and reproducible comparison among the three recurrent architectures evaluated in this study [7,9]. This study first formalizes the multi-target forecasting problem and provide an overview of the two datasets used—a local microgrid time series and the OPSD Germany dataset [18]. It then detail the preprocessing steps applied uniformly across all models, including missing-value handling, feature scaling, and the 24 h input $\rightarrow$ 12 h forecast sliding-window sequence construction [1,6]. In the following sections, This Study present the three neural network architectures evaluated in this study: a standard single-layer LSTM as the basic baseline, a stacked bidirectional LSTM (BiLSTM), and the proposed AttentionBiLSTM that incorporates additive (Bahdanau-style) attention on top of the bidirectional recurrent layers [10,12,17]. All three models are trained under exactly the same conditions, including the use of a combined loss function defined as $MSE + 0.5 \times MAE$ and identical optimizer hyperparameters (Adam with the same learning rate schedule) [15]. To guarantee a strictly fair comparison, every architecture receives the identical set of input features, undergoes the same data normalization and preprocessing steps, and is subject to the same early-stopping criterion based on validation loss [13,20]. Finally,

model performance and robustness are assessed through a 5-fold rolling-origin (time-series) cross-validation scheme on both datasets. This procedure expands the training window progressively while keeping future data unseen, thereby preventing information leakage and closely replicating operational forecasting settings [7,8]. Model performance is assessed using RMSE, nRMSE, MAPE, sMAPE, and the standard deviation of RMSE across folds, consistent with established practice in energy forecasting research [5,11]. In addition, paired t -tests are conducted on fold-wise RMSE values to evaluate the statistical significance of performance differences among the models [16].

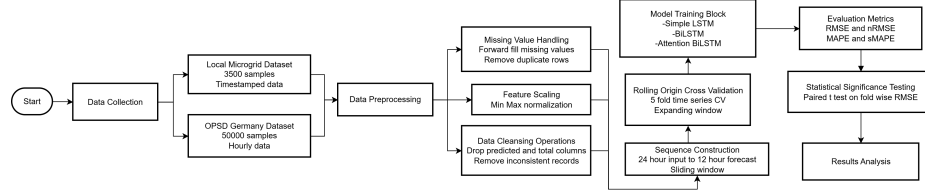


Fig. 1: Overall workflow of the proposed multi-target forecasting framework.

3.1 Forecasting Task

Let $y_t^{(sol)}$, $y_t^{(wind)}$ and $y_t^{(load)}$ denote solar PV output, wind power output and grid load demand at time step t , respectively. At each time step this also observe a feature vector $\mathbf{z}_t \in \mathbb{R}^F$ that contains these targets together with additional engineered variables. The goal is to learn a function

$$f_\theta: \mathbb{R}^{N_{in} \times F} \rightarrow \mathbb{R}^{N_{out} \times 3}, \quad (1)$$

which maps the past $N_{in} = 24$ hours of features to the next $N_{out} = 12$ hours of the three target series. For each reference time t the input and output blocks are defined as

$$\mathbf{X}_t = [\mathbf{z}_{t-N_{in}+1}, \dots, \mathbf{z}_t], \quad \mathbf{Y}_t = \begin{bmatrix} y_{t+1}^{(sol)} & y_{t+1}^{(wind)} & y_{t+1}^{(load)} \\ \vdots & \vdots & \vdots \\ y_{t+N_{out}}^{(sol)} & y_{t+N_{out}}^{(wind)} & y_{t+N_{out}}^{(load)} \end{bmatrix}, \quad (2)$$

and the model seeks to produce $\hat{\mathbf{Y}}_t = f_\theta(\mathbf{X}_t)$ that approximates $\mathbf{Y}_t$.

3.2 Datasets

This study considers two datasets with different spatial scales: a local microgrid dataset and the national-level OPSD Germany dataset.

Local Microgrid Dataset The local microgrid dataset consists of historical time-series measurements of solar generation, wind power, electrical load, and related operational variables for a single microgrid. To prevent data leakage,

columns associated with precomputed forecasts or cumulative energy values are removed. Missing values are handled through forward filling, with any remaining incomplete rows discarded. After preprocessing and the addition of lagged features (Section 3.3), the resulting dataset characteristics are summarized in Table 1.

OPSD Germany Dataset The OPSD Germany dataset provides hourly national-level time series of grid load, solar PV generation, and wind power generation. Relevant variables are extracted from the official OPSD time-series collection and combined with temporal features and the preprocessing steps described earlier. The final dataset composition is summarized in Table 1.

Table 1: Dataset description after preprocessing and feature construction.

Dataset Name	Source	Samples	Features	Targets	Horizon
Local Microgrid Dataset	Local microgrid	3,378	22	Solar, Wind, Load	12-step
OPSD Germany Dataset	Open Power System Data	50,225	10	Solar, Wind, Load	12-step

3.3 Feature Engineering and Scaling

Temporal features are derived from timestamps by extracting the hour of day and day of week for each dataset. To better capture load dynamics, lagged load features at 1, 24, and 168 hours are constructed as

$$\text{load_lag_1}_t = y_{t-1}^{(\text{load})}, \quad \text{load_lag_24}_t = y_{t-24}^{(\text{load})}, \quad \text{load_lag_168}_t = y_{t-168}^{(\text{load})}. \quad (3)$$

Rows for which these lagged values are undefined are discarded.

Temporal periodicity is encoded using standard sinusoidal transformations:

$$\text{hour_sin}_t = \sin\left(2\pi \frac{\text{hour_of_day}_t}{24}\right), \quad \text{hour_cos}_t = \cos\left(2\pi \frac{\text{hour_of_day}_t}{24}\right), \quad (4)$$

$$\text{dow_sin}_t = \sin\left(2\pi \frac{\text{day_of_week}_t}{7}\right), \quad \text{dow_cos}_t = \cos\left(2\pi \frac{\text{day_of_week}_t}{7}\right). \quad (5)$$

All numerical features, including targets and engineered variables, are scaled to the range $[0, 1]$ using a MinMax scaler fitted on the training portion of each dataset and subsequently applied to the corresponding validation and test segments.

3.4 Sequence Construction

After preprocessing and scaling, each dataset is transformed into supervised learning pairs using a sliding window. For every valid time index t we construct $\mathbf{X}_t$ and $\mathbf{Y}_t$ as defined above, yielding a collection of $(\mathbf{X}_i, \mathbf{Y}_i)$ pairs that can be fed to sequence-to-sequence models. This construction is performed independently for each dataset and for each fold in the rolling-origin validation protocol.

3.5 Model Architectures

To evaluate three recurrent architectures: a SimpleLSTM baseline, a stacked BiLSTM and the proposed AttentionBiLSTM. All models are implemented in TensorFlow/Keras and trained under identical hyper-parameters.

SimpleLSTM The SimpleLSTM model uses a single LSTM layer containing 100 units followed by ReLU activation. The final hidden state is passed through a dense layer whose output is reshaped into the desired $N_{\text{out}} \times 3$ forecast matrix:

$$\mathbf{h} = \text{LSTM}_{100}(\mathbf{X}), \quad \mathbf{o} = W_o \mathbf{h} + \mathbf{b}_o, \quad \hat{\mathbf{Y}} = \text{reshape}(\mathbf{o}, N_{\text{out}}, 3). \quad (6)$$

BiLSTM The BiLSTM model consists of two stacked bidirectional LSTM layers, each having 150 units with ReLU activation. The first layer returns the full output sequence, while the second layer returns only the final hidden state.

$$\mathbf{H}^{(1)} = \text{BiLSTM}_{150}^{\text{seq}}(\mathbf{X}), \quad \mathbf{h}^{(2)} = \text{BiLSTM}_{150}^{\text{final}}(\mathbf{H}^{(1)}), \quad (7)$$

followed by a dense layer and reshape as in the SimpleLSTM architecture.

AttentionBiLSTM The proposed AttentionBiLSTM shares the same stacked BiLSTM backbone but introduces a Bahdanau-style additive attention mechanism applied to the sequence of hidden states. The two bidirectional LSTM layers produce a sequence $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_T]$, where $T = N_{\text{in}}$. We define the query as the last time step $\mathbf{q} = \mathbf{h}_T$ and compute an alignment score for each position τ :

$$e_\tau = \mathbf{v}^\top \tanh(W_q \mathbf{q} + W_h \mathbf{h}_\tau + \mathbf{b}), \quad (8)$$

with W_q , W_h , $\mathbf{v}$ and $\mathbf{b}$ being trainable parameters. Attention weights are obtained through a softmax,

$$\alpha_\tau = \frac{\exp(e_\tau)}{\sum_{k=1}^T \exp(e_k)}, \quad \sum_{\tau=1}^T \alpha_\tau = 1, \quad (9)$$

and the context vector is given by

$$\mathbf{c} = \sum_{\tau=1}^T \alpha_\tau \mathbf{h}_\tau. \quad (10)$$

The forecast is generated by a dense projection followed by a reshape:

$$\mathbf{o} = W_o \mathbf{c} + \mathbf{b}_o, \quad \hat{\mathbf{Y}} = \text{reshape}(\mathbf{o}, N_{\text{out}}, 3). \quad (11)$$

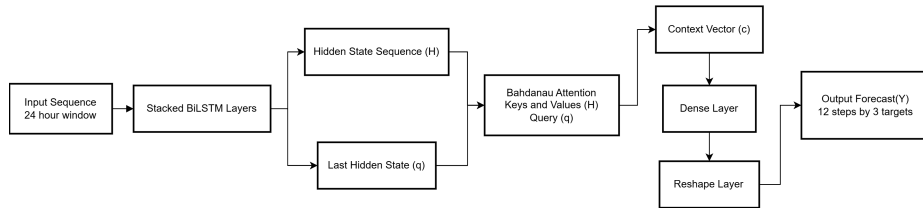


Fig. 2: Compact AttentionBiLSTM architecture with Bahdanau additive attention.

3.6 Loss Function and Optimization

To be compared fairly, all three models are trained with the same hybrid loss, which combines the mean squared error (MSE) and mean absolute error (MAE):

$$\mathcal{L}_{\text{hybrid}}(\mathbf{Y}, \hat{\mathbf{Y}}) = \text{MSE}(\mathbf{Y}, \hat{\mathbf{Y}}) + 0.5 \text{MAE}(\mathbf{Y}, \hat{\mathbf{Y}}), \quad (12)$$

where MSE and MAE are computed across all forecast horizons and all three targets. All models are trained with the Adam optimizer using the default learning rate of 0.001, a fixed batch size of 64, and up to 100 epochs. Early stopping halts training if validation loss does not improve for 10 straight epochs (patience = 10). A ReduceLROnPlateau scheduler also lowers the learning rate by a factor of 0.5 when validation loss plateaus for 5 epochs. No model-specific hyper-parameter tuning is performed; this deliberate choice keeps the comparison focused on architectural effects rather than aggressive optimization.

3.7 Rolling-Origin Evaluation Protocol

For each dataset, approximately 70% of the time series is used as the initial training region, while the remaining 30% is kept aside for evaluation.

The evaluation region is then divided into $K = 5$ consecutive folds. In fold k , the training window includes the initial 70% plus all previous folds, and the current fold is used only as the test window. No shuffling is performed at any point during the process.

Within each fold, all the data up to the end of the test window are first transformed into input-output pairs $(\mathbf{X}_i, \mathbf{Y}_i)$. For every one of the five folds, the training and test segments are separated chronologically using a fixed time-based split point.

This approach ensures that, in every fold, all three competing models—SimpleLSTM, BiLSTM, and AttentionBiLSTM—are trained and evaluated on exactly the same sequences. As a result, differences in performance come only from the model architectures themselves, with no bias introduced by varying data partitioning [7,8].

3.8 Evaluation Metrics

Model performance is evaluated using standard regression metrics for short-term energy forecasting [19]. For each target (solar PV generation, wind power production, and electrical load) and for each cross-validation fold, we report the root mean squared error (RMSE) and mean absolute error (MAE) to measure absolute accuracy. The normalized RMSE (nRMSE) is calculated by dividing RMSE by the mean absolute ground-truth value over the test period, allowing comparison across variables with different scales [19]. Relative errors are assessed

with the mean absolute percentage error (MAPE), computed only when $|y| > 10^{-3}$, along with the symmetric mean absolute percentage error (sMAPE). Model robustness is assessed using the standard deviation of RMSE across the five folds. All results correspond to averages over valid folds. In addition, fold-wise RMSE values are subjected to paired t -tests to assess the statistical significance of performance differences between AttentionBiLSTM and the baseline models (BiLSTM and SimpleLSTM).

4 Results and Discussion

This study assesses the performance of the three recurrent models — SimpleLSTM, BiLSTM, and AttentionBiLSTM — on two benchmark datasets: the local microgrid dataset and the OPSD Germany dataset [18]. The evaluation is based on RMSE, nRMSE, MAPE, sMAPE, and RMSE standard deviation, computed under a 5-fold rolling-origin cross-validation protocol. Among these metrics, nRMSE is used in Figures 3 and 4 to provide a scale-independent visual comparison across targets and models.

For the local microgrid dataset, all three models were evaluated on solar PV, wind power, and load forecasting. The results show that for grid load demand, the AttentionBiLSTM obtains the lowest average RMSE (127.53) and nRMSE (0.4821) among the three models. Paired t -tests on fold-wise RMSE indicate that differences between AttentionBiLSTM and the baselines are not statistically significant at the 5% level for any target. For example, for load the p -values are 0.1006 (vs. BiLSTM) and 0.0588 (vs. SimpleLSTM), as shown in Table 4.

For the OPSD Germany dataset, the three models were again evaluated jointly on solar PV, wind power, and load. The results show that all three models achieve very low error for grid load demand, with sMAPE values around 2%. For wind power, the SimpleLSTM clearly outperforms the other two architectures in terms of RMSE, nRMSE, and percentage errors. The paired t -tests reveal that for OPSD wind power, the SimpleLSTM performs significantly better than the AttentionBiLSTM ($p \approx 0.0280$), while differences for solar and load fail to achieve statistical significance at the 5% level. Overall, the findings indicate that model effectiveness is closely tied to the characteristics of each forecasting target. AttentionBiLSTM provides stronger stability for highly variable microgrid load, while SimpleLSTM excels on smoother national-scale wind patterns. These observations establish clear performance baselines for future multi-target energy forecasting research.

MAPE is computed only where $|y| > 10^{-3}$ to avoid instability due to near-zero solar values. RMSE values are reported in native units as shown in Table 2

OPSD solar values include many zero or near-zero readings (nighttime), causing higher MAPE and sMAPE variability as shown in Table 3

Practical implications of statistical significance: Statistical significance indicates that observed performance differences are consistent across rolling-origin folds rather than due to random data splits (see Table 4). Even marginal but statistically

reliable improvements can enhance robustness and decision confidence in short-term operational forecasting, whereas non-significant differences suggest that added model complexity may not yield practical benefits.

Computational complexity and training cost: All models were trained under identical hardware conditions on Google Colab for up to 100 epochs, with total training time ranging from approximately 45 minutes to 1 hour. SimpleLSTM has the lowest training cost, followed by BiLSTM, while AttentionBiLSTM incurs the highest cost due to additional attention computations; inference time remains low and suitable for real-time applications.

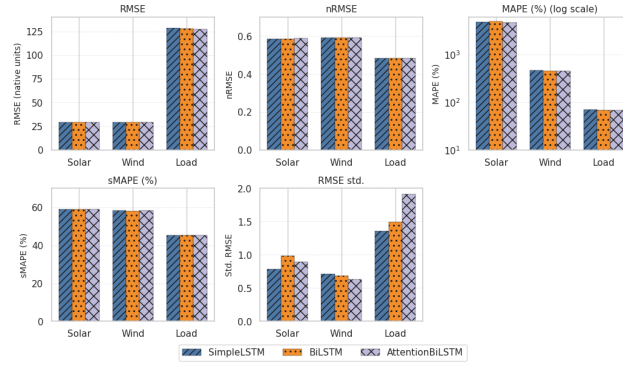


Fig. 3: Comparison of forecasting performance for solar, wind, and load on the local microgrid dataset.

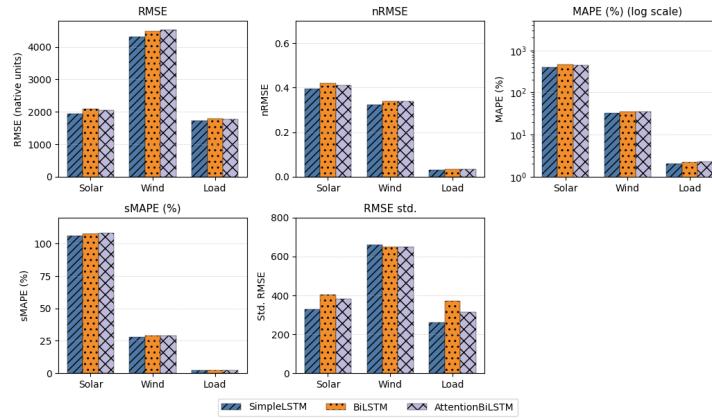


Fig. 4: Comparison of forecasting performance for solar, wind, and load on the OPSP Germany dataset.

Table 2: Local microgrid dataset: average performance across folds.

Target	Model	RMSE	nRMSE	MAPE (%)	sMAPE (%)	RMSE std.
Solar	SimpleLSTM	29.2480	0.5859	4743.71	58.86	0.7855
	BiLSTM	29.2032	0.5849	4875.35	59.01	0.9827
	AttentionBiLSTM	29.3023	0.5870	4668.76	59.17	0.8921
Wind	SimpleLSTM	29.0111	0.5907	461.65	58.23	0.7171
	BiLSTM	29.0210	0.5909	452.08	58.21	0.6880
	AttentionBiLSTM	29.0082	0.5906	452.20	58.32	0.6336
Load	SimpleLSTM	128.3009	0.4850	69.28	45.30	1.3597
	BiLSTM	128.0591	0.4841	68.41	45.41	1.4959
	AttentionBiLSTM	127.5262	0.4821	68.37	45.32	1.9147

Table 3: OPSD Germany (DE) dataset: average performance across 5-fold rolling-origin cross-validation.

Target	Model	RMSE	nRMSE	MAPE (%)	sMAPE (%)	RMSE std.
Solar	SimpleLSTM	1932.9995	0.3958	397.68	105.82	329.2947
	BiLSTM	2083.2925	0.4206	466.18	107.39	404.5997
	AttentionBiLSTM	2057.3802	0.4155	437.73	108.32	383.6136
Wind	SimpleLSTM	4314.8488	0.3242	32.03	27.90	660.8261
	BiLSTM	4495.6712	0.3383	35.03	28.86	650.6628
	AttentionBiLSTM	4520.3987	0.3399	35.59	29.03	649.1186
Load	SimpleLSTM	1725.6803	0.0316	2.07	2.05	262.1069
	BiLSTM	1784.8083	0.0326	2.17	2.16	371.4459
	AttentionBiLSTM	1773.0157	0.0324	2.29	2.27	313.0438

Table 4: Paired t -test p -values comparing AttentionBiLSTM with BiLSTM and SimpleLSTM across datasets and targets (RMSE across five folds).

Local Microgrid			OPSD Germany		
Target	Comparison	p -value	Target	Comparison	p -value
Solar	Att. vs. SimpleLSTM	0.4016	Solar	Att. vs. SimpleLSTM	0.1155
	Att. vs. BiLSTM	0.1415		Att. vs. BiLSTM	0.6081
Wind	Att. vs. SimpleLSTM	0.9551	Wind	Att. vs. SimpleLSTM	0.0280
	Att. vs. BiLSTM	0.7245		Att. vs. BiLSTM	0.6101
Load	Att. vs. SimpleLSTM	0.0588	Load	Att. vs. SimpleLSTM	0.4269
	Att. vs. BiLSTM	0.1006		Att. vs. BiLSTM	0.8275

Bold indicates statistical significance at the 5% level.

5 Conclusion and Future Work

This study presented a fair, multi-model and multi-dataset comparison of SimpleLSTM, BiLSTM, and AttentionBiLSTM for short-term forecasting of solar

PV, wind power, and electrical load under a unified preprocessing, training, and evaluation pipeline. Results show that national-scale load forecasting on the OPSD dataset achieves very low error across all models, whereas solar and wind prediction—particularly on the local microgrid—remains challenging due to sparse and highly variable signals. The AttentionBiLSTM provides modest gains in stability for certain targets, but is not consistently superior; notably, SimpleLSTM achieves a statistically significant RMSE advantage for OPSD wind power. Overall, attention mechanisms enhance robustness in specific scenarios, while simpler architectures remain competitive when strong temporal patterns are present.

Future work will explore Transformer-based architectures and probabilistic forecasting approaches for uncertainty quantification, along with enhanced explainability techniques to further analyze model behavior.

Author Contributions

Guggilla Deepu contributed to the conceptualization of the study, dataset preprocessing, model implementation, experimental evaluation, and drafting of the manuscript. Dontha Madhusudhana Rao helped design the methodology, supervised the overall research, validated the results, and carried out a critical review and revision of the manuscript. Dokuri Harjith Reddy contributed by performing data analysis, running the experimental evaluations, and helping interpret the results. Mukundu Sai Varun contributed to the literature review, visualization of results, and manuscript editing. All authors read and approved the final manuscript.

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